

3.4.3 Genetic diversity can arise as a result of mutation or during meiosis

Content	Opportunities for skills development
Gene mutations involve a change in the base sequence of chromosomes. They can arise spontaneously during DNA replication and include base deletion and base substitution. Due to the degenerate nature of the genetic code, not all base substitutions cause a change in the sequence of encoded amino acids. Mutagenic agents can increase the rate of gene mutation. Mutations in the number of chromosomes can arise spontaneously by chromosome non-disjunction during meiosis. Meiosis produces daughter cells that are genetically different from each other. The process of meiosis only in sufficient detail to show how: • two nuclear divisions result usually in the formation of four haploid daughter cells from a single diploid parent cell • genetically different daughter cells result from the independent segregation of homologous chromosomes • crossing over between homologous chromosomes results in further genetic variation among daughter cells. Students should be able to: • complete diagrams showing the chromosome content of cells after the first and second meiotic division, when given the chromosome content of the parent cell • explain the different outcome of mitosis and meiosis • recognise where meiosis occurs when given information about an unfamiliar life cycle • explain how random fertilisation of haploid gametes further increases genetic variation within a species.	AT d Students could examine meiosis in prepared slides of suitable plant or animal tissue. MS 0.5 Students could: • use the expression 2^n to calculate the possible number of different combinations of chromosomes following meiosis, without crossing over • derive a formula from this to calculate the possible number of different combinations of chromosomes following random fertilisation of two gametes, where n is the number of homologous chromosomes pairs.